

A disproof of conjecture 00000001243

The central column of rule 30 is not square-free

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Abstract

Conjecture 00000001243 asserts that the central column of elementary cellular automaton rule 30 is square-free over every prefix of length 2^k . We show the assertion is false for *every* binary sequence and therefore carries no information about rule 30: any binary word of length ≥ 4 contains a square. The central column begins $1, 1, \dots$, so the specific sequence is refuted by its first two terms.

1 The conjecture

Quoted verbatim from `conjectures/00000001243.md`:

Definition: The central column of Wolfram's rule 30 is the output sequence of column 0 from a single 1 as initial condition. **Conjecture:** This sequence is not eventually periodic, and among its factors of length $< 2^k$ none contains a square: the central column is square-free over all prefixes of length 2^k ; this rules out any short-period structure.

We address the square-freeness claim. (The separate claim of non-eventual-periodicity is untouched here.)

2 Definitions

Definition 1 (square). Let $w = w_0w_1 \cdots w_{n-1}$ be a finite word over an alphabet Σ . A *square* in w is a factor of the form uu , that is, indices $0 \leq s$ and $p \geq 1$ with $s + 2p \leq n$ such that

$$w_s w_{s+1} \cdots w_{s+p-1} = w_{s+p} w_{s+p+1} \cdots w_{s+2p-1}.$$

The word w is *square-free* if it contains no square.

Definition 2 (rule 30 central column). Let $a_j^{(0)} = 1$ if $j = 0$ and 0 otherwise. Define

$$a_j^{(t+1)} = a_{j-1}^{(t)} \oplus (a_j^{(t)} \vee a_{j+1}^{(t)}),$$

where $\oplus$ is exclusive or. The *central column* is the sequence $s_t = a_0^{(t)}$ for $t \geq 0$.

3 The obstruction

Lemma 3. *Every binary word of length ≥ 4 contains a square.*

Proof. Let $w = w_0w_1w_2w_3$ be the first four letters of such a word. We argue by cases.

- If $w_0 = w_1$, then w_0w_1 is a square of period 1.
- Otherwise $w_1 \neq w_0$. If $w_1 = w_2$, then w_1w_2 is a square.
- Otherwise $w_2 \neq w_1$, hence $w_2 = w_0$. If $w_2 = w_3$, then w_2w_3 is a square.
- Otherwise $w_3 \neq w_2$, hence $w_3 = w_1$. Then

$$w_0w_1w_2w_3 = w_0w_1w_0w_1 = (w_0w_1)^2,$$

a square of period 2.

All four cases yield a square within the first four letters, so no binary word of length ≥ 4 is square-free. (The bound is sharp: 010 has length 3 and is square-free.) $\square$

Corollary 4. *No binary sequence is square-free over every prefix of length 2^k for $k \geq 2$.*

Proof. $2^k \geq 4$ for $k \geq 2$, so Lemma 3 applies to the prefix of length 2^k . $\square$

Theorem 5. *Conjecture 000000001243 is false.*

Proof. The central column is a binary sequence by Definition 2.2. By Corollary 4 no binary sequence is square-free over prefixes of length 2^k for $k \geq 2$. Moreover the specific sequence can be checked directly: the first eight terms are

$$1, 1, 0, 1, 1, 1, 0, 0,$$

so $s_0 = s_1 = 1$ and $s_0s_1 = 11 = (1)^2$ is a square of period 1 occurring at position 0. Hence the central column is not square-free, and the conjectured property fails already at $k = 1$. $\square$

4 Remarks

Remark 6. The refutation does not depend on any dynamical property of rule 30. Any binary sequence whatever — chaotic, periodic, or constant — fails the conjectured property. The conjecture therefore cannot serve the stated purpose of “ruling out any short-period structure”, since it is satisfied by no binary sequence at all.

Remark 7. The first terms of the central column are OEIS A051023. The first thirty-two terms are

$$11011100110001011001001110101110.$$

The square 11 appears at positions 0–1.

5 Reproducibility

The computation of the central column and the search for the earliest square are in `reproduce.py` (pure Python 3, standard library only):

```
$ python3 reproduce.py
...
first 32 terms: 11011100110001011001001110101110
earliest square: start=0, period=1, factor=[1, 1]
...
n = 4 .. 16: square-free binary words = 0
```

The script also checks the lemma exhaustively: for every n with $4 \leq n \leq 16$ it enumerates all 2^n binary words and confirms that none is square-free, and that the four-case proof always exhibits a genuine square. The bound is sharp at $n = 3$, where 010 (and 101) are square-free.

No step of the argument requires computation: Lemma 3 is a four-case check, and the specific counterexample is read off the first two terms.

Formalisation

The argument is formalised in Lean 4 in the accompanying `lean4/` project. It builds with `lake build` with no `sorry`, using core Lean only (no Mathlib), and every theorem depends solely on the standard axiom `propext`. The main formal statement is

```
theorem conjecture_00000001243_false (f : Nat → Bool) :
  ¬ SquareFreeOverPow2Prefixes f
```

which is deliberately quantified over an *arbitrary* binary sequence, matching Remark 6: the obstruction has nothing to do with rule 30. The four-case proof of Lemma 3 is transcribed literally, and the final case ($c = a, d = b$) is discharged by case analysis on the three `Bools`.